

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Safety, Pharmacokinetics and Preliminary Anti-Tumor Activity of Englumafusp Alfa in Combination With Obinutuzumab and in Combination With Glofitamab Following a Pre-Treatment Dose of Obinutuzumab in Participants With Relapsed/Refractory B-Cell Non-Hodgkin's Lymphoma **Short Title/Acronym:** BP41072 Study **Protocol Number:** BP41072 (EudraCT: 2019-000416-28) **NCT Number:** NCT04077723 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Englumafusp alfa (R07227166) in combination with Obinutuzumab (Gpt) and Glofitamab (Tocilizumab may be utilized/referenced in safety management) **Phase of Development:** Phase I/II (PHASE1, PHASE2) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, dose-limiting toxicities (DLTs), Overall Response Rate (ORR), and to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose of englumafusp alfa in combination with obinutuzumab and glofitamab. **Secondary Objectives:** To evaluate specific pharmacokinetics (PK) parameters including AUC, Cmax, and Cmin of the combination regimen.

2.2 Background & Rationale To address the need for durable, "off-the-shelf" treatments for relapsed/refractory aggressive B-cell non-Hodgkin lymphoma (r/r B-NHL). Glofitamab (a CD20xCD3 T-cell engager) has proven efficacy, but combining it with englumafusp alfa (a CD19/4-1BBL costimulatory bispecific antibody-like fusion protein) aims to synergistically deepen and prolong anti-tumor responses, providing an alternative to second-generation CAR T-cell therapy with a positive benefit-risk profile.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Dose-escalation [Part I and II] and Dose-expansion [Part III]) **Control Type:** Single-arm (per cohort) **Blinding:** Open-label **Randomization:** N/A (Sequential assignment/Dose escalation)

3.2 Planned Enrollment & Duration Target **SN\$:** 498 participants **Number of Sites:** Multicenter/Global **Estimated Study Start:** 2019 **Trial Status:** Recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. History or status of a histologically-confirmed hematological malignancy expected to express CD19 and CD20. 2. Relapse after or failure to respond to at least one prior treatment regimen (Part I and II: no available treatment options expected to prolong survival; Part III: relapsed after or failed to respond to only one prior systemic treatment regimen). 3. Measurable target lesion (≥ 1.5 cm) in its largest dimension by computed tomography (CT) scan. 4. Able and willing to provide a fresh biopsy from a safely accessible site. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 (or ≤ 2 for some participants in Part III). 6. Life expectancy of ≥ 12 weeks. 7. Adverse events from prior anti-cancer therapy resolved to Grade ≤ 1 . 8. Adequate liver, hematological, and renal function. 9. Negative test results for HIV, acute or chronic hepatitis B virus (HBV), and hepatitis C virus (HCV). 10. Strict contraception/abstinence adherence: Women of childbearing potential and male participants must use highly effective contraception during treatment and for extended periods post-treatment (e.g., 18 months after obinutuzumab, 3.5 months after englumafusp alfa, 2 months after glofitamab, or 3 months after tocilizumab [ATC: L04AC07], whichever is longer).

4.2 Exclusion Criteria 1. Circulating lymphoma cells, defined by out of range absolute lymphocyte count (ALC) or abnormal cells in peripheral blood. 2. Acute bacterial, viral, fungal, mycobacterial, or other active infections, or positive blood culture within 72 hours prior to obinutuzumab infusion. 3. Pregnant or breast-feeding or intending to become pregnant. 4. Prior treatment with systemic immunotherapeutic agents within 4 weeks or five half-lives. 5. Prior solid organ transplantation or allogeneic stem cell transplant, or autologous stem cell transplant within 100 days prior to obinutuzumab infusion. 6. Current or past history of central nervous system (CNS) lymphoma and CNS disease. 7. Evidence of significant, uncontrolled concomitant diseases (e.g., diabetes mellitus, pulmonary disorders, autoimmune diseases). 8. Major surgery or significant traumatic injury < 28 days prior to Gpt infusion. 9. Another invasive malignancy in the last 2 years or significant cardiovascular disease. 10. Administration of a live, attenuated vaccine within 4 weeks before Gpt infusion. 11. Receipt of systemic immunosuppressive medications within two weeks prior to Gpt (stable corticosteroid treatment ≤ 25 mg/day of prednisone equivalent is permitted).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Pre-treatment:** A fixed single dose of obinutuzumab (1000mg) administered up to 7 days prior to treatment initiation with englumafusp alfa and glofitamab. * **Combination:** Glofitamab administered via step-up

dosing in Cycles 1 and 2. Englumafusp alfa is started one week later. From Cycle 3 Day 1 onwards, both agents are given on the same day every 3 weeks (Q3W). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Fixed treatment duration.

5.2 Concomitant & Prohibited Therapy Permitted: Corticosteroid treatment ≤ 25 mg/day of prednisone (or equivalent) if on a stable dose for at least 2 weeks prior; inhaled/topical steroids. Tocilizumab may be used per standard protocols for AE management. **Prohibited:** Systemic immunosuppressive medications (e.g., cyclophosphamide, azathioprine, methotrexate, thalidomide, anti-TNF agents) within 2 weeks prior to obinutuzumab pre-treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, Tumor Imaging, Baseline Labs, Viral Screening
Pre-Treatment	Day -7 to -1	Obinutuzumab pre-treatment IV infusion to mitigate tumor burden/CRS
Cycle 1-2	Step-up Dosing	Glofitamab step-up dosing; close monitoring for CRS
Cycle 3-12	Every 3 Weeks (Q3W)	Simultaneous IV infusion of Englumafusp alfa + Glofitamab, PK/PD tracking, Efficacy Assessment
End of Study	Follow-up	Response durability tracking, Long-term safety follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

1. Nature and frequency of dose-limiting toxicities (DLTs) to establish the Maximum Tolerated Dose (MTD).
2. Proportion of Participants with Adverse Events (AEs): Incidence, nature, and severity of AEs graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) v5.0.
3. Overall Response Rate (ORR).

7.2 Secondary Endpoints

1. Total exposure (area under the concentration time curve [AUC]) of englumafusp alfa in combination with obinutuzumab and glofitamab.
2. Maximum serum concentration (peak concentration, C_{max}) of englumafusp alfa in combination with obinutuzumab and glofitamab.
3. Minimum serum concentration (trough concentration, C_{min}) of englumafusp alfa in combination with obinutuzumab and glofitamab.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard AE reporting with strict protocols for evaluating Cytokine Release Syndrome (CRS). Graded via NCI CTCAE v5.0. **Serious Adverse Events (SAEs):** Routine monitoring and reporting per regulatory guidelines. **Data**

Monitoring Committee (DMC): Utilized a modified Continual Reassessment Method (mCRM) with Escalation With Overdose Control (EWOC) principle for dose escalation.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety-evaluable populations for AE/DLT incidence; Efficacy-evaluable populations for ORR. **Sample Size Justification:** Target enrollment of 498 participants utilizing a Bayesian adaptive design (mCRM EWOC) for dose-escalation cohorts to identify MTD efficiently. **Handling of Missing Data:** Standard protocol-defined handling for non-evaluable responses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Intensive inpatient/outpatient monitoring during Cycle 1/Cycle 2 step-up dosing specifically for prompt CRS management. **Audit Trail:** Electronic Case Report Form (eCRF) tracking with centralized independent review of radiological tumor responses.

11. Study Identifiers & Governance

Primary ID: BP41072 **Registry Number:** NCT04077723 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** BP41072 **Official Regulatory Title:** An Open-Label, Phase I/II Study to Evaluate the Safety, Pharmacokinetics and Preliminary Anti-Tumor Activity of Englumafusp Alfa (R07227166, A CD19 Targeted 4-1BB Ligand) in Combination With Obinutuzumab and in Combination With Glofitamab Following a Pre-treatment Dose of Obinutuzumab Administered in Participants With Relapsed/Refractory B-Cell Non-Hodgkin's Lymphoma

12. Clinical Framework (Oncology Specific)

Primary Condition: Lymphoma, Non-Hodgkin (Relapsed/Refractory B-Cell Non-Hodgkin's Lymphoma [r/r B-NHL]) **Diagnosis Threshold:** Refractory to or relapsed after at least one prior line of systemic therapy **Medical Methodology:** Bispecific antibody combination therapy (T-cell engager + costimulatory fusion protein) **Optimization Target:** Maximum Tolerated Dose (MTD) and Overall Response Rate (ORR)

13. Study Procedures & Methodology

Management Pathway: Step-up dosing clinical pathway to mitigate Cytokine Release Syndrome (CRS) **Education Protocol (HCP):** Site training on early recognition and management of CRS and neurologic toxicities **Education Protocol (Patient):** Education on symptom recognition (e.g., fever, hypotension) and reporting post-infusion **Quality Audit Mechanism:** Centralized PK/PD lab evaluation and centralized radiological review

14. Observation & Follow-Up Schedule

Total Duration: Follow-up extending past treatment completion (incorporating specific follow-up times up to 18 months for pregnancy prevention protocols depending on therapy administered) **Onsite Visit Cadence:** Pre-treatment phase, intensive Step-up days (Cycles 1-2), followed by Every 3 Weeks (Q3W) **Remote Monitoring Frequency:** Routine telemedicine/symptom check-ins during off-weeks **Checklist Review Interval:** Disease response assessment every few cycles via CT scans

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				